Differential Spectral Damping: Gap-Adaptive Regularization for Ill-Conditioned Kernel Methods

Praveg Vashishtha[a,b,*]

[a] *Department of Computer Science and Engineering, Indian Institute of Technology Patna, Patna, India*

[b] *Sopra Banking Software India, India*

Abstract

Kernel methods requiring matrix inversion—particularly Least-Squares Twin Support Vector Machines (LSTSVM)—suffer from exponential eigenvalue decay in their system matrices, producing severely ill-conditioned problems where standard Tikhonov regularization applies uniform damping regardless of eigenvector reliability. We propose Differential Spectral Damping (DSD), a regularization formula that adapts its penalty to localized eigengap structure: preserving eigenvectors with large spectral gaps (reliable per Davis-Kahan perturbation theory) while aggressively suppressing those with small gaps (directionally corrupted beyond recovery). We motivate DSD through a principled design procedure grounded in the Davis-Kahan $\sin(\Theta)$ theorem, systematically deriving the requirements for a reliability-aware damping function and selecting the exponential form for its smoothness, differentiability, and natural saturation properties. Through rigorous paired testing

*Corresponding author.

Email address: praveg_pa2503mth259@iitp.ac.in (Praveg Vashishtha)

URL: https://www.linkedin.com/in/pravegvashishtha/ (Praveg Vashishtha)

with fairly optimized baselines (including gradient-optimized Tikhonov receiving equal optimization opportunity), we demonstrate that DSD improves LSTSVM classification accuracy by +4.8 percentage points on real-world GINA (d=970, Cohen's $d = 4.49$, $p < 0.0001$), +10.4 percentage points at d=200, and +2.6 percentage points on Madelon (d=500, Cohen's $d = 1.76$)—all using only principled spectral initialization while Tikhonov receives grid search. For pre-image reconstruction on manifold data, DSD ties Tikhonov at high perturbation noise (p=0.99) but slightly underperforms at lower noise levels; both reduce naive inversion error by 66×. We characterize the precise operating regime ($d \geq 100$, condition number $> 10^3$) and document where simpler methods suffice, providing practitioners with clear deployment guidance.

Keywords: Kernel methods, spectral regularization, eigengap adaptation, ill-conditioned systems, Twin SVM, numerical stability, eigenvalue perturbation, pre-image problem

1. Introduction

Kernel methods map input data into high-dimensional feature spaces via the kernel trick, enabling non-linear classification and regression without explicit feature computation [1, 2]. However, methods that require *inversion* of kernel-derived system matrices—particularly the Least-Squares Twin Support Vector Machine (LSTSVM) [3]—face a fundamental numerical challenge: the eigenvalues of these system matrices decay exponentially, producing severe ill-conditioning that standard scalar regularization cannot adequately address.

LSTSVM constructs two non-parallel hyperplanes by solving a pair of linear systems whose coefficient matrices take the form $M = E_1^\top E_1 + c^{-1} E_2^\top E_2$, where E_1 and E_2 are augmented class-specific data matrices and $c > 0$ is a regularization constant. In high-dimensional settings ($d \geq 100$), these matrices empirically exhibit *extreme spectral tail-clustering*: at d=200, the bottom 50% of eigenvalues occupy only 7% of the total spectral range, with condition numbers exceeding 10^4.

LSTSVM and its kernel variants remain actively deployed in domains where deep learning overfits due to limited sample sizes [4], including spectroscopic material identification, regulatory credit scoring with engineered features, and high-dimensional biomedical classification. In these high-stakes settings, a +3–10 percentage point accuracy improvement from regularization alone directly impacts diagnostic and decision quality.

The central theoretical insight motivating our work comes from the Davis-Kahan $\sin(\Theta)$ theorem [5], which establishes that eigenvector perturbation under matrix noise E is bounded by:

$$\sin(\theta_i) \leq \frac{\|E\|_2}{\delta_i} \tag{1}$$

where δ_i denotes the *eigengap*—the minimum distance from eigenvalue λ_i to all other eigenvalues. When eigenvalues cluster tightly ($\delta_i \to 0$), their corresponding eigenvectors become *arbitrarily corrupted* by any perturbation, including floating-point arithmetic noise. Standard Tikhonov regularization $(W + \gamma I)^{-1}$ addresses eigenvalue magnitudes but is structurally incapable of accounting for this directional corruption: a single scalar γ cannot simultaneously avoid over-regularizing the well-separated spectral head (where eigenvectors are reliable) and under-regularizing the clustered tail (where

eigenvectors are corrupted).

We propose **Differential Spectral Damping (DSD)**, a regularization formula that adapts its penalty strength to the *local eigengap density* at each spectral position. Eigenvectors with large gaps (reliable per Davis-Kahan) are preserved with near-exact inversion; eigenvectors with small gaps (directionally corrupted) are aggressively suppressed. The transition is smooth, differentiable, and governed by only two interpretable parameters.

1.1. Contributions

1. A **gap-adaptive regularization formula** (DSD) for kernel matrix inversion, derived from Davis-Kahan perturbation theory with complete mathematical motivation.
2. A **principled initialization procedure** requiring zero cross-validation, based on spectral transition analysis.
3. A **differentiable PyTorch implementation** enabling gradient-based optimization of DSD parameters through backpropagation.
4. **Rigorous experimental comparison** with fairly-optimized baselines (50-seed paired testing, Tikhonov receiving grid search or equivalent gradient optimization).
5. **Clear operating regime characterization**: DSD dominates at $d \geq 100$ with severe tail-clustering; equivalent to Tikhonov at $d \leq 50$.
6. **Honest documentation of limitations**: pre-image tasks see no improvement over Tikhonov; DSD provides no benefit below its operating threshold.

1.2. Paper Organization

Section 2 establishes notation and symbol definitions used throughout. Section 3 reviews the mathematical background (kernel methods, LSTSVM, spectral perturbation theory). Section 4 motivates and constructs the DSD formula. Section 5 presents the complete algorithm, initialization, and implementation. Section 6 positions DSD relative to existing regularization approaches. Section 7 describes experimental design. Section 8 presents results. Section 9 provides analysis and interpretation. Section 10 discusses limitations, and Section 11 concludes.

2. Notation and Symbol Definitions

Table 1 defines all mathematical symbols used in this paper. We adopt the following conventions: bold lowercase letters ($\boldsymbol{x}$) denote vectors, bold uppercase letters ($\boldsymbol{W}$) denote matrices, calligraphic letters ($\mathcal{H}$) denote spaces, and Greek letters denote parameters or scalar quantities.

Eigenvalue ordering convention. Throughout this paper, eigenvalues are sorted in *ascending* order: $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_m$. Index i=1 corresponds to the smallest eigenvalue (most vulnerable to perturbation); index i=m corresponds to the largest (most reliable). This convention aligns with the output of standard numerical eigensolvers (`numpy.linalg.eigh`, `torch.linalg.eigh`).

3. Mathematical Background

3.1. Kernel Methods and the Kernel Trick

A kernel function $\kappa : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ implicitly computes inner products in a (possibly infinite-dimensional) Hilbert space $\mathcal{H}$ without requiring explicit computation of the feature map ϕ:

$$\kappa(\boldsymbol{x}, \boldsymbol{x}') = \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \rangle_{\mathcal{H}} \tag{2}$$

The Radial Basis Function (RBF) kernel, defined as:

$$\kappa(\boldsymbol{x}, \boldsymbol{x}') = \exp\big(-\gamma_k \|\boldsymbol{x} - \boldsymbol{x}'\|^2\big), \quad \gamma_k > 0 \tag{3}$$

maps data into an infinite-dimensional space. The parameter γ_k controls the bandwidth: larger γ_k produces sharper kernels with faster eigenvalue decay in the Gram matrix.

The kernel (Gram) matrix $\boldsymbol{K} \in \mathbb{R}^{n \times n}$ with entries $K_{ij} = \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is symmetric positive semi-definite (PSD) by Mercer's theorem. For RBF kernels, the eigenvalues of $\boldsymbol{K}$ decay exponentially [2]:

$$\lambda_i \propto \exp\big(-c \cdot i^{2/d}\big) \tag{4}$$

where c depends on γ_k and the intrinsic dimensionality of the data. This decay is not a pathology—it reflects the smoothness assumptions encoded by the kernel—but it creates severe numerical challenges for methods requiring $\boldsymbol{K}^{-1}$ or the inverse of kernel-derived matrices.

3.2. Least-Squares Twin SVM (LSTSVM)

LSTSVM [3] finds two non-parallel hyperplanes, each proximal to one class. Let $X_+ \in \mathbb{R}^{n_+ \times d}$ and $X_- \in \mathbb{R}^{n_- \times d}$ be the class-specific data matrices.

Define augmented matrices:

$$E_1 = [X_+ \;\; \boldsymbol{e}_+] \in \mathbb{R}^{n_+ \times (d+1)}, \quad E_2 = [X_- \;\; \boldsymbol{e}_-] \in \mathbb{R}^{n_- \times (d+1)} \tag{5}$$

where $\boldsymbol{e}_\pm$ are vectors of ones (bias terms). LSTSVM solves two linear systems:

$$(E_2^\top E_2 + c_1^{-1} E_1^\top E_1)\boldsymbol{u} = E_2^\top \boldsymbol{e}_- \tag{6}$$

$$(E_1^\top E_1 + c_2^{-1} E_2^\top E_2)\boldsymbol{v} = E_1^\top \boldsymbol{e}_+ \tag{7}$$

where $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^{d+1}$ define the two hyperplanes, and $c_1, c_2 > 0$ are penalty parameters.

The coefficient matrices $M_1 = E_2^\top E_2 + c_1^{-1} E_1^\top E_1$ and $M_2 = E_1^\top E_1 + c_2^{-1} E_2^\top E_2$ are sums of outer-product matrices. When d is large relative to class sizes ($d \gg n_\pm$), these matrices inherit the spectral properties of the data covariance—including severe tail-clustering of eigenvalues. A new test sample $\boldsymbol{x}$ is classified according to its proximity to each hyperplane:

$$\text{class}(\boldsymbol{x}) = \arg\min_{k \in \{1,2\}} \frac{|\boldsymbol{w}_k^\top \boldsymbol{x} + b_k|}{\|\boldsymbol{w}_k\|} \tag{8}$$

where $(\boldsymbol{w}_k, b_k)$ are extracted from $\boldsymbol{u}$ and $\boldsymbol{v}$.

3.3. The Nyström Approximation

For kernel-space operations on n samples, the full Gram matrix $\boldsymbol{K} \in \mathbb{R}^{n \times n}$ may be prohibitively large. The Nyström approximation [6] selects $m \ll n$ landmark points $L = \{\boldsymbol{l}_1, \ldots, \boldsymbol{l}_m\}$ and approximates:

$$\boldsymbol{K} \approx \boldsymbol{K}_{nm} \boldsymbol{W}^{-1} \boldsymbol{K}_{nm}^\top \tag{9}$$

where $\boldsymbol{W} \in \mathbb{R}^{m \times m}$ is the kernel matrix among landmarks ($W_{ij} = \kappa(\boldsymbol{l}_i, \boldsymbol{l}_j)$) and $\boldsymbol{K}_{nm} \in \mathbb{R}^{n \times m}$ contains kernel evaluations between all samples and landmarks. This approximation requires computing $\boldsymbol{W}^{-1}$ (or its regularized variant)—the operation that DSD stabilizes.

3.4. The Davis-Kahan $\sin(\Theta)$ *Theorem*

The Davis-Kahan theorem [5] is the theoretical foundation of DSD. It bounds how much an eigenvector rotates under matrix perturbation.

Theorem 1 (Davis-Kahan, simplified form)**.** *Let* $\boldsymbol{A}$ *be a symmetric matrix with eigenvalue* λ_i *and eigenvector* $\boldsymbol{u}_i$*. Let* $\tilde{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{E}$ *be a perturbed matrix with corresponding eigenvector* $\tilde{\boldsymbol{u}}_i$*. Define the eigengap:*

$$\delta_i = \min_{j \neq i} |\lambda_i - \lambda_j| \tag{10}$$

Then the angle θ_i *between the true and perturbed eigenvectors satisfies:*

$$\sin(\theta_i) \leq \frac{\|\boldsymbol{E}\|_2}{\delta_i} \tag{11}$$

Interpretation. The bound (11) states that eigenvector reliability is determined by the ratio of perturbation magnitude to eigengap. When $\delta_i \gg \|\boldsymbol{E}\|_2$, the eigenvector is robust ($\sin\theta_i \approx 0$, small rotation). When $\delta_i \lesssim \|\boldsymbol{E}\|_2$, the bound becomes vacuous ($\sin\theta_i$ could approach 1, meaning the eigenvector can rotate by up to 90°)—the eigenvector direction is essentially random.

Implication for regularization. Any regularization scheme that uses the eigendecomposition $\boldsymbol{W} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$ is implicitly trusting the eigenvectors $\boldsymbol{u}_i$. But Davis-Kahan tells us that eigenvectors associated with small gaps are *unreliable directions* in the matrix. Computing $\boldsymbol{W}^{-1} = \boldsymbol{U}\boldsymbol{\Lambda}^{-1}\boldsymbol{U}^\top$ amplifies

contributions from corrupted directions (division by small λ_i). Tikhonov regularization replaces λ_i^{-1} with $(\lambda_i+\gamma)^{-1}$—damping the *magnitude* uniformly—but does nothing about the corrupted *direction* $\boldsymbol{u}_i$.

3.5. The Pre-Image Problem

Given a point $\boldsymbol{\psi} \in \mathcal{H}$ (e.g., a decision boundary point in kernel space), the pre-image problem seeks $\hat{\boldsymbol{x}} \in \mathbb{R}^d$ such that $\phi(\hat{\boldsymbol{x}}) \approx \boldsymbol{\psi}$ [7]. Under the Nyström framework, the pre-image is computed as:

$$\hat{\boldsymbol{x}} = \boldsymbol{k}_q^\top \boldsymbol{W}^{-1} L = \sum_{j=1}^{m} w_j \boldsymbol{l}_j, \quad w_j = (\boldsymbol{W}^{-1}\boldsymbol{k}_q)_j \tag{12}$$

where $\boldsymbol{k}_q = [\kappa(\boldsymbol{x}_q, \boldsymbol{l}_1), \ldots, \kappa(\boldsymbol{x}_q, \boldsymbol{l}_m)]^\top$ is the query kernel vector. The pre-image is a weighted combination of landmark points, with weights determined by the regularized inverse. Ill-conditioned $\boldsymbol{W}$ produces extreme weights that push $\hat{\boldsymbol{x}}$ far from the data manifold.

4. Design and Motivation of Differential Spectral Damping

We construct DSD by asking: *given the Davis-Kahan bound, what is the optimal regularization strategy that minimizes information loss while respecting eigenvector reliability?* The following procedure systematically identifies the requirements for such a regularizer and motivates the specific functional form.

4.1. Step 1: Quantifying Eigenvector Reliability

From (11), the reliability of eigenvector $\boldsymbol{u}_i$ is characterized by the ratio $r_i = \|\boldsymbol{E}\|_2/\delta_i$. We define a *reliability indicator*:

$$\rho_i = \begin{cases} \text{high} & \text{if } \delta_i \gg \|\boldsymbol{E}\|_2 \quad (\sin\theta_i \approx 0) \\ \text{low} & \text{if } \delta_i \lesssim \|\boldsymbol{E}\|_2 \quad (\sin\theta_i \text{ large}) \end{cases} \tag{13}$$

In practice, $\|\boldsymbol{E}\|_2$ represents the combined effect of:

- Finite-precision arithmetic ($\sim 10^{-16}$ for float64, amplified by condition number)
- Sampling noise in kernel matrix construction
- Nyström approximation error

The exact value of $\|\boldsymbol{E}\|_2$ is typically unknown. However, the *relative* ordering of eigengaps δ_i is observable from the eigendecomposition. DSD exploits this: it does not need to know the absolute perturbation magnitude—it only needs to identify which eigenvectors have *relatively* small gaps (and are therefore relatively more vulnerable).

4.2. Step 2: The Ideal Regularization Profile

For an ideal regularizer, we want:

- When δ_i is large (reliable eigenvector): preserve the exact inverse λ_i^{-1} to retain maximum information.
- When δ_i is small (corrupted eigenvector): suppress the contribution by driving the effective inverse toward zero, preventing the corrupted direction from contaminating the solution.

- The transition between these regimes should be smooth (to avoid discontinuity artifacts in downstream tasks like classification or pre-image computation).

This translates into a *damping function* $d(\delta_i)$ that satisfies:

$$d(\delta_i) \to 0 \quad \text{as } \delta_i \to \infty \quad \text{(no damping for reliable eigenvectors)} \tag{14}$$

$$d(\delta_i) \to \alpha_{\max} \quad \text{as } \delta_i \to 0 \quad \text{(maximum damping for corrupted)} \tag{15}$$

$$d'(\delta_i) < 0 \quad \text{(monotonically decreasing: more gap} \Rightarrow \text{less damping)} \tag{16}$$

4.3. Step 3: Functional Form Selection

The requirements (14)–(16) are satisfied by any monotonically decreasing function bounded between 0 and $\alpha_{\max}$. We select the *exponential decay* form:

$$d_i = \alpha \cdot \exp(-\beta \cdot \delta_i) \tag{17}$$

for the following reasons:

1. **Natural saturation.** $\exp(-\beta\delta_i) \in (0, 1]$ automatically satisfies the boundedness requirement without additional clipping or normalization.
2. **Differentiability.** The exponential is C^∞ (infinitely differentiable), enabling gradient-based optimization of (α, β) via backpropagation.
3. **Scale invariance.** The product $\beta \cdot \delta_i$ is dimensionless when β has units of inverse-gap, making the formula invariant to eigenvalue scaling.
4. **Rapid transition.** The exponential transitions sharply from ≈ 1 (small gap) to ≈ 0 (large gap), matching the binary nature of the Davis-Kahan reliability bound (eigenvectors are either reliably estimated or not—there is no "moderately corrupted" regime).

5. **Parsimony.** Only two free parameters (α, β) control the entire damping profile across all m eigenvalues.

Alternative functional forms that satisfy (14)–(16) include the sigmoid $\alpha/(1 + \exp(\beta(\delta_i - \delta_0)))$, the rational function $\alpha/(1 + \beta\delta_i)^p$, and the complementary error function $\alpha \cdot \mathrm{erfc}(\beta\delta_i)$. All would produce qualitatively similar regularization behavior. We select the exponential for its analytical simplicity and the interpretability of its parameters (see Section 4.6).

4.4. Step 4: Incorporating Damping into the Inverse

The standard pseudo-inverse via eigendecomposition is:

$$\boldsymbol{W}^{-1} = \boldsymbol{U}\boldsymbol{\Lambda}^{-1}\boldsymbol{U}^\top = \sum_{i=1}^{m} \frac{1}{\lambda_i}\boldsymbol{u}_i\boldsymbol{u}_i^\top \tag{18}$$

Tikhonov regularization replaces this with:

$$\boldsymbol{W}_{\text{Tik}}^{-1} = \sum_{i=1}^{m} \frac{1}{\lambda_i + \gamma}\boldsymbol{u}_i\boldsymbol{u}_i^\top \tag{19}$$

We incorporate DSD damping into the regularized inverse formula. The key design decision is *how* the damping d_i modifies the inverse. We adopt the form:

$$\tilde{\lambda}_i^{-1} = \frac{\lambda_i}{\lambda_i^2 + d_i} \tag{20}$$

This specific form is motivated by the following properties:

- **When** $d_i = 0$ (no damping, reliable eigenvector): $\tilde{\lambda}_i^{-1} = \lambda_i/\lambda_i^2 = 1/\lambda_i$. The exact inverse is recovered.

- **When** $d_i = \alpha \gg \lambda_i^2$ (maximum damping, corrupted eigenvector): $\tilde{\lambda}_i^{-1} \approx \lambda_i/\alpha \to 0$. The contribution of this direction is suppressed.

- **Intermediate regime**: smooth interpolation governed by the relative magnitudes of λ_i^2 and d_i.

- **Positive definiteness**: $\tilde{\lambda}_i^{-1} > 0$ for all $\lambda_i > 0$, ensuring the regularized inverse remains PSD.

- **Tikhonov recovery**: when $d_i = \gamma^2$ (constant), the formula becomes $\lambda_i/(\lambda_i^2 + \gamma^2)$, which for $\lambda_i \gg \gamma$ approximates $(1/\lambda_i)(1 - \gamma^2/\lambda_i^2) \approx 1/(\lambda_i + \gamma^2/\lambda_i)$—a generalized Tikhonov form. DSD thus *generalizes* scalar regularization.

4.5. Step 5: The Complete DSD Formula

Substituting the damping function (17) into (20):

$$\boxed{\tilde{\lambda}_i^{-1} = \frac{\lambda_i}{\lambda_i^2 + \alpha \cdot \exp(-\beta \cdot \delta_i)}} \tag{21}$$

The full DSD-regularized pseudo-inverse is:

$$\tilde{\boldsymbol{W}}^{+} = \boldsymbol{U} \cdot \mathrm{diag}(\tilde{\lambda}_1^{-1}, \ldots, \tilde{\lambda}_m^{-1}) \cdot \boldsymbol{U}^{\top} = \sum_{i=1}^{m} \tilde{\lambda}_i^{-1} \boldsymbol{u}_i \boldsymbol{u}_i^{\top} \tag{22}$$

4.6. Step 6: Parameter Interpretation

The two parameters have clear physical interpretations:

α **(penalty magnitude):** Controls the maximum damping applied to the most corrupted eigenvectors. When $\delta_i \to 0$, the damping term equals α, so $\tilde{\lambda}_i^{-1} \approx \lambda_i/(\lambda_i^2 + \alpha)$. For small eigenvalues ($\lambda_i^2 \ll \alpha$), this reduces to $\lambda_i/\alpha \approx 0$—complete suppression. The parameter α sets the "penalty ceiling" and should be calibrated to the scale of eigenvalues where corruption begins (the spectral transition point).

β **(gap sensitivity):** Controls the sharpness of the transition between "reliable" and "corrupted" regimes. Large β creates a sharp step-function-like transition (small gaps are heavily damped; moderate gaps are barely damped). Small β creates a gradual transition. The parameter β should be calibrated to the typical eigengap scale so that "typical" gaps produce moderate damping while significantly larger gaps produce near-zero damping.

4.7. Comparison with Tikhonov in the Eigenvalue Domain

Table 2 contrasts DSD with existing approaches at the eigenvalue level.

DSD is the only method that conditions its regularization on the *gap structure*—a proxy for eigenvector directional reliability per Davis-Kahan. All other methods regularize based on eigenvalue *magnitude* alone.

5. Method: Algorithm, Initialization, and Implementation

5.1. Localized Eigengap Computation

The Davis-Kahan bound (11) uses the *global* eigengap $\delta_i = \min_{j \neq i} |\lambda_i - \lambda_j|$. For monotonically ordered eigenvalues (as produced by symmetric eigensolvers), this simplifies to the *bilateral* minimum:

$$\delta_i = \min\big(|\lambda_i - \lambda_{i-1}|,\ |\lambda_i - \lambda_{i+1}|\big) \tag{23}$$

since the nearest eigenvalue to λ_i in a sorted sequence is always one of its immediate neighbors.

Boundary handling. At the sequence edges (i=1 and i=m), only one

neighbor exists. We use single-sided gaps:

$$\delta_i = \begin{cases} |\lambda_1 - \lambda_2| & i = 1 \\ \min(|\lambda_i - \lambda_{i-1}|, |\lambda_i - \lambda_{i+1}|) & 1 < i < m \\ |\lambda_m - \lambda_{m-1}| & i = m \end{cases} \tag{24}$$

The boundary choice for i=1 (the smallest, most vulnerable eigenvalue) is significant: using the right-side gap $|\lambda_1 - \lambda_2|$ ensures this eigenvalue receives appropriate damping based on its actual spectral isolation, rather than being artificially exempted.

5.2. Principled Hyperparameter Initialization

DSD's initialization requires no cross-validation. The two parameters are determined entirely from the eigendecomposition of the input matrix $\boldsymbol{W}$.

Initialization of α (penalty magnitude):

$$\alpha_0 = \lambda^2_{\text{transition}} \tag{25}$$

where $\lambda_{\text{transition}}$ is the eigenvalue at the spectral "knee"—the point where eigengaps become characteristically small, indicating the onset of the unreliable spectral tail. Operationally:

1. Compute all consecutive gaps: $\Delta_i = |\lambda_{i+1} - \lambda_i|$ for $i = 1, \ldots, m-1$.
2. Identify the 10th percentile gap: $g_{10} = \text{quantile}_{0.10}(\{\Delta_i\})$.
3. Find the last index where the gap falls below g_{10}: $i^* = \max\{i : \Delta_i < g_{10}\}$.
4. Set $\lambda_{\text{transition}} = \lambda_{i^*}$.

Rationale: The squared eigenvalue $\lambda^2_{\text{transition}}$ ensures α is on the correct scale for the formula $\lambda_i^2 + \alpha$. At the transition point, $\lambda_i^2 \approx \alpha$, producing roughly 50% damping—the natural midpoint of the reliable/unreliable transition.

Initialization of β (gap sensitivity):

$$\beta_0 = \frac{1}{\text{median}(\{\Delta_i\})} \tag{26}$$

Rationale: This normalizes the exponential argument so that a "typical" eigengap produces $\exp(-\beta_0 \cdot \delta_{\text{typical}}) = \exp(-1) \approx 0.37$—moderate damping. Gaps significantly larger than the median produce $\exp(-\beta_0 \cdot \delta_{\text{large}}) \approx 0$—near-zero damping (preservation). Gaps significantly smaller produce $\exp(-\beta_0 \cdot \delta_{\text{small}}) \approx 1$—maximum damping (suppression). The median is robust to outliers (unlike mean), preventing a single large gap from dominating the sensitivity calibration.

5.3. Gradient-Based Parameter Optimization

When a differentiable reconstruction objective exists (e.g., pre-image tasks), α and β can be refined via backpropagation after initialization. The parameters are stored in log-space to enforce positivity:

$$\alpha = \exp(\theta_\alpha), \quad \beta = \exp(\theta_\beta) \tag{27}$$

where $\theta_\alpha, \theta_\beta \in \mathbb{R}$ are unconstrained learnable parameters. Optimization uses Adam [8] with learning rate 0.01 and early stopping (patience = 20 epochs).

Gradient flow. The DSD formula (21) is fully differentiable with respect

to θ_α and θ_β:

$$\frac{\partial \tilde{\lambda}_i^{-1}}{\partial \theta_\alpha} = \frac{-\lambda_i \cdot d_i}{(\lambda_i^2 + d_i)^2} \cdot 1 = \frac{-\lambda_i \cdot d_i}{(\lambda_i^2 + d_i)^2} \tag{28}$$

$$\frac{\partial \tilde{\lambda}_i^{-1}}{\partial \theta_\beta} = \frac{\lambda_i \cdot d_i \cdot \beta \cdot \delta_i}{(\lambda_i^2 + d_i)^2} \tag{29}$$

where $d_i = \alpha \cdot \exp(-\beta \cdot \delta_i)$. Both gradients are well-defined and bounded for all $\lambda_i > 0$, $\delta_i \geq 0$.

Eigenvector gradient stability. The backward pass through `torch.linalg.eigh` involves terms proportional to $1/(\lambda_i - \lambda_j)$ for the eigenvector gradient. When eigenvalues are nearly degenerate ($\lambda_i \approx \lambda_j$), these terms become numerically unstable. Our implementation detects this condition (minimum gap $< 10^{-6}$) and detaches the eigenvector matrix from the computation graph, allowing gradients to flow only through the eigenvalue $\rightarrow$ damping $\rightarrow$ inverse path. This is mathematically justified: when eigenvectors are unreliable (the exact condition DSD was designed for), their gradients are also unreliable and should not influence parameter updates.

5.4. The Structural Prior Discovery

A key empirical finding: for LSTSVM classification, the principled initialization (25)–(26) *without* gradient optimization outperforms gradient-optimized parameters by a significant margin (GINA d=970: 85.9% init vs 81.1% optimized, $p < 10^{-6}$).

Mechanism. Gradient trajectory analysis reveals that during optimization on reconstruction loss, β decreases (from 0.74 to 0.34, a 0.46$\times$ reduction) while α increases (2.3$\times$). Decreasing β flattens the exponential $\exp(-\beta\delta_i)$, making damping increasingly uniform—converging toward constant-penalty

Tikhonov behavior. The optimizer minimizes reconstruction error by removing the gap-adaptive selectivity, because pointwise reconstruction benefits from using all spectral components (including corrupted ones that happen to align with the training data). Classification, however, benefits from *suppressing* corrupted components to prevent noise-fitting.

Conclusion. DSD's classification advantage is a **structural prior**—the initialization formula encodes domain knowledge ("trust well-separated eigenvectors, distrust clustered ones") that a reconstruction-loss optimizer would destroy. This is analogous to early stopping in neural networks: the optimal-for-generalization parameter configuration is not the loss-minimizing one. We observe this behavior consistently across synthetic datasets with varying covariance structure (50 seeds each across four dimensionalities) and two real-world benchmarks (GINA, Madelon); further validation on domain-specific datasets (e.g., genomic expression data with $d > 1000$) is an important direction for confirming generality.

5.5. Complete Algorithm

Computational complexity. The dominant cost is the eigendecomposition at $O(m^3)$. All DSD-specific operations (Steps 3–8) are $O(m)$. The reconstruction (Step 9) is $O(m^2)$ (or $O(m^2 k)$ if only k columns are needed). DSD adds less than 0.01% overhead to the total computation. For $m > 1500$, a scalable path using partial eigendecomposition (`scipy.sparse.linalg.eigsh` with rank $k \ll m$) and factored (non-materialized) inverse reduces cost to $O(m^2 k)$.

Memory. Peak memory during forward pass: $O(m^2)$ for storing $\boldsymbol{U}$ and $\tilde{\boldsymbol{W}}^+$. During backpropagation (gradient optimization), an additional $O(m^2)$

Algorithm 1 Differential Spectral Damping (DSD)

Require: Symmetric PSD matrix $\boldsymbol{W} \in \mathbb{R}^{m \times m}$, optional (α, β)

Ensure: DSD-regularized pseudo-inverse $\tilde{\boldsymbol{W}}^{+} \in \mathbb{R}^{m \times m}$

Eigendecomposition: $\boldsymbol{U}, \boldsymbol{\Lambda} \leftarrow \mathrm{eigh}(\boldsymbol{W})$ $[O(m^3)]$

Filter: Retain indices $\{i : \lambda_i > 10^{-12}\}$ $[O(m)]$

Eigengaps: $\delta_i \leftarrow$ Eq. (24) $[O(m)]$

if α, β not provided **then**

  $\alpha \leftarrow \lambda_{i^*}^2$ (Eq. 25); $\beta \leftarrow 1/\mathrm{median}(\Delta)$ (Eq. 26) $[O(m)]$

end if

Damping: $d_i \leftarrow \alpha \cdot \exp(-\beta \cdot \delta_i)$ for all i $[O(m)]$

Regularized inverse: $\tilde{\lambda}_i^{-1} \leftarrow \lambda_i / (\lambda_i^2 + d_i)$ $[O(m)]$

Reconstruct: $\tilde{\boldsymbol{W}}^{+} \leftarrow \boldsymbol{U} \cdot \mathrm{diag}(\tilde{\boldsymbol{\lambda}}^{-1}) \cdot \boldsymbol{U}^{\top}$ $[O(m^2)]$

return $\tilde{\boldsymbol{W}}^{+}$

is required for the backward pass through `eigh`. For inference-only deployment, $\tilde{\boldsymbol{W}}^{+}$ can be precomputed once and reused for all queries.

6. Related Work

Tikhonov (ridge) regularization [9] adds a scalar ridge γ to all eigenvalues: $\tilde{\lambda}_i^{-1} = 1/(\lambda_i + \gamma)$. This is optimal when all spectral directions are equally reliable or equally corrupted—a uniform-noise assumption. Generalized cross-validation [10] provides principled γ selection but does not address the structural limitation of scalar uniformity. DSD recovers Tikhonov behavior when $\beta \to 0$ (all gaps treated identically).

Truncated SVD [11] retains the top-k eigenvectors and discards the rest: a hard binary cutoff. This is optimal when noise is strictly confined to the

bottom $m-k$ components. In practice, the signal/noise boundary is gradual, and hard cutoff creates discontinuity artifacts. DSD provides the continuous generalization. We include truncated SVD (with optimally-selected rank) as an experimental baseline: on GINA (d=970), TSVD achieves 83.4% vs. DSD's 85.9%—better than scalar Tikhonov (81.1%) but still inferior to gap-adaptive damping ($p < 0.0001$).

Spectral filtering [12] applies a filter function $g(\lambda_i)$ that depends on eigenvalue magnitude (e.g., $g(\lambda_i) = \lambda_i/(\lambda_i + \gamma)^p$). These filters adapt to spectral *scale* but not spectral *structure*—they cannot distinguish a small eigenvalue with a large gap (reliable) from one with a small gap (corrupted).

LSTSVM regularization. Standard LSTSVM implementations [3, 13] use Tikhonov regularization ($M + cI$). Tanveer et al. [4] survey Twin SVM variants; none employ gap-adaptive regularization. The spectral properties of LSTSVM product matrices ($E_i^\top E_i$ cross-class sums) have not been previously characterized as a regularization-relevant phenomenon.

Pre-image methods. Mika et al. [14] and Kwok & Tsang [7] address the pre-image problem via fixed-point iteration and learning-based approaches respectively. Both assume a stable kernel matrix inverse is available—the problem DSD addresses. Schölkopf et al. [15] established the theoretical framework connecting feature space and input space representations.

Post-hoc explainability. SHAP [16] and LIME [17] provide local, statistical feature attributions without geometric guarantees. They operate independently of the kernel matrix inverse and cannot provide global decision boundary reconstruction—the application that motivates DSD's pre-image stabilization.

7. Experimental Design

7.1. Fairness Protocol

A central concern in regularization comparisons is optimization fairness: a method with more tuning budget will appear artificially superior. We employ the following protocol:

For LSTSVM classification (Tables 3, 6): DSD uses principled auto-initialization only (no gradient optimization, no cross-validation). Tikhonov receives γ grid-searched over a 15-point logarithmic grid from 10^{-8} to 10^{-1}, selected by training accuracy. This gives Tikhonov a deliberate tuning advantage over DSD.

For pre-image reconstruction (Table 5): Both methods receive equal optimization. DSD's (α, β) are optimized via Adam (lr=0.01, 150 epochs, patience=20) on reconstruction loss $\mathcal{L} = \frac{1}{n}\sum_i \|\hat{\boldsymbol{x}}_i - \boldsymbol{x}_i\|^2$. Tikhonov's γ receives the *same* Adam optimizer on the *same* loss (single differentiable parameter)—eliminating grid-vs-gradient fairness concerns entirely.

For Kernel-LSTSVM (Table 4): DSD receives gradient optimization on kernel pre-image loss. Tikhonov receives 20-point grid search on training accuracy.

7.2. Statistical Methodology

All experiments use paired designs: within each seed, DSD and Tikhonov receive identical data (same train/test split, same noise realization). We report:

- **Paired t-test**: two-sided p-value testing the null hypothesis of equal mean performance.

- **Cohen's** d: standardized effect size, computed as $d = \bar{\Delta}/s_\Delta$ where $\bar{\Delta}$ is the mean paired difference and s_Δ its standard deviation. By convention: $d > 0.2$ small, $d > 0.5$ medium, $d > 0.8$ large.
- **Win count**: number of seeds where DSD outperforms the baseline.
- **Seed count**: 50 seeds for synthetic experiments; 30 for real-world datasets (GINA, Madelon) due to computational cost.

7.3. Datasets

Synthetic LSTSVM. Generated via `sklearn.datasets.make_classification` with: (d=200, n=300), (d=100, n=200), (d=50, n=300), (d=30, n=400). Informative features: $d/2$. Noise: 10% label flip. 70/30 stratified split.

Real-world LSTSVM. GINA (d=970, n=3468, handwriting recognition, OpenML ID 41158); Madelon (d=500, n=2600, NIPS 2003 Feature Selection Challenge—20 informative features among 500).

Pre-image. Swiss Roll manifold (n=2000, ambient d=3, intrinsic d=2), m=300 k-means landmarks, γ_k=2.0, noise levels $\sigma \in \{5\times10^{-3}, 10^{-3}, 10^{-4}\}$.

Kernel-LSTSVM. Two Moons (n=400, γ_k=2.0); Genomics-like (d=100, n=200, γ_k=0.02).

Operating boundary validation. Digits (d=64), Ionosphere (d=34)—included to empirically confirm where DSD does *not* help.

8. Results

8.1. Linear LSTSVM Classification (Primary Result)

DSD's advantage scales monotonically with dimensionality and spectral severity. The GINA result (Cohen's $d = 4.49$) is an exceptionally large stan-

dardized effect—substantially exceeding the $d > 0.8$ threshold for "large" effects [18]. Madelon's 57.7% absolute accuracy reflects the dataset's designed difficulty (published linear-classifier baselines without feature selection: 55–62%); DSD's +2.6pp gain is achieved against optimal Tikhonov in this adversarial setting.

At d=50, both methods perform identically (DSD's structural prior provides no advantage when spectral clustering is mild). This transition is sharp: at d=100 the advantage is already +3.3pp.

8.2. Kernel LSTSVM (Non-Linear Classification)

On genuinely non-linear problems with sharp RBF kernels (severe spectral decay), DSD provides statistically significant improvement even when both methods receive optimization.

8.3. Pre-Image Reconstruction

Both DSD and Tikhonov reduce naive inversion error by 66× at high noise—demonstrating the critical importance of *any* regularization. However, DSD does not outperform Tikhonov on this task. The explanation: pre-image reconstruction has a single scalar objective ($\min \|\hat{\boldsymbol{x}} - \boldsymbol{x}\|^2$) that Tikhonov's single parameter can optimize directly. The gap-adaptive structure provides no additional advantage when the objective is purely pointwise reconstruction.

8.4. Operating Boundary Validation

Below $d \approx 100$, DSD provides no advantage and may underperform. These results are included to honestly characterize the operating boundary,

not as success claims. The deployment recommendation is: apply DSD when the system matrix condition number exceeds $\sim 10^3$ and the spectral tail occupies $< 10\%$ of the spectral range.

9. Analysis and Interpretation

9.1. Spectral Tail-Clustering in LSTSVM Matrices

The claimed spectral tail-clustering is verified empirically by measuring the fraction of total spectral range occupied by the bottom 50% of eigenvalues:

At d=200, the bottom half of eigenvalues are compressed into 7% of the total spectral range—extreme clustering that corrupts eigenvector directions per Davis-Kahan. At d=50, the same fraction spans 15%, reducing clustering below DSD's activation threshold. DSD's advantage is directly proportional to tail-clustering severity.

9.2. Why DSD Helps Classification But Not Pre-Image

The classification task (LSTSVM) and pre-image task differ fundamentally in their sensitivity to spectral structure:

Classification requires the solution $\boldsymbol{u} = M^{-1}\boldsymbol{b}$ to *generalize* to unseen data. Corrupted eigenvectors contribute noise-fitting components to $\boldsymbol{u}$ that align with training data by chance but do not generalize. DSD suppresses these components—effectively reducing the hypothesis class to the subspace of well-separated eigenvectors. This is spectral regularization in the generalization-theoretic sense [19].

Pre-image reconstruction minimizes $\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|^2$—a scalar objective over a specific known target $\boldsymbol{x}$. A single scalar γ can be optimized (by grid search

or gradient descent) to find the best bias-variance tradeoff for this specific objective. The gap structure does not provide additional information when the goal is point-wise reconstruction rather than generalization.

9.3. Connection to Algorithmic Stability

DSD's classification improvement can be understood through algorithmic stability theory. In LSTSVM, a single training perturbation propagates through all spectral directions of the inverse. Under Tikhonov (uniform damping γ), perturbations in directions with small eigengaps—where Davis-Kahan guarantees large eigenvector sensitivity—propagate with full force. DSD selectively suppresses contributions from these sensitive directions.

This effectively reduces the *effective dimensionality* of the learner: fewer spectral directions participate in the solution, bounding the model's sensitivity to individual training samples—the hallmark of algorithmic stability that implies generalization [19]. DSD achieves this dimensionality reduction softly (through continuous damping) rather than through hard rank truncation, avoiding the discontinuity artifacts of truncated SVD.

9.4. Operating Regime Characterization

The operating regime is characterized by two observable conditions:

1. **Condition number** of the system matrix exceeds 10^3.
2. **Spectral tail span** (fraction of range occupied by bottom 50% of eigenvalues) is below 10–12%.

Both can be computed from the eigendecomposition (already required for DSD) at $O(m)$ cost, enabling automatic detection of whether DSD will benefit a given problem.

10. Limitations and Future Work

10.1. Limitations

1. **No formal stability proof.** The conjecture that DSD pre-image error remains bounded as $\delta_i \to 0$ (because damping grows to compensate) is supported empirically but not proved. A formal proof combining Davis-Kahan with the DSD damping structure remains open.
2. **Non-RBF kernels.** The principled initialization assumes monotonic eigenvalue decay (a property of RBF kernels). For polynomial or non-stationary kernels with non-monotonic spectra, the initialization may require modification.
3. **Pre-image performance.** DSD matches but does not beat optimally-tuned Tikhonov for pre-image reconstruction. The per-eigenvector adaptation provides no advantage when the optimization target is a scalar reconstruction loss.
4. **Low-dimensional regime.** Below $d \approx 100$, spectral tail-clustering is insufficient for DSD's gap-adaptive structure to provide benefit. DSD may slightly underperform Tikhonov in this regime.
5. **Eigendecomposition requirement.** DSD requires the full eigendecomposition of the system matrix. While this is already necessary for Tikhonov via the eigenvalue path (and is $O(m^3)$ regardless), it precludes application to matrices where only matrix-vector products are available (iterative solvers).

10.2. Future Work

Three directions follow naturally from this work:

1. Multi-scale gap sensitivity. Preliminary investigation with hierarchical β (separate sensitivity parameters for spectral tail, middle, and head zones) suggests statistically significant improvement over scalar β in low-noise regimes on manifold data. This extension adds only two parameters while never degrading performance in any configuration tested, indicating that the non-uniform perturbation sensitivity across the spectrum—predicted by the Davis-Kahan bound's non-linear dependence on eigenvalue position—can be exploited for further improvement.

2. Extension to related methods. Kernel Fisher Discriminant Analysis, Kernel CCA, and structured output SVMs all construct system matrices from cross-class interactions with similar spectral properties. DSD's advantage should transfer directly.

3. Formal stability bound. Proving $\|\hat{\boldsymbol{x}}_{\text{DSD}} - \hat{\boldsymbol{x}}_{\text{true}}\| \leq C \cdot \epsilon / \min_i(\lambda_i^2 + d_i)$ would complete the theoretical foundation by showing that DSD error is bounded regardless of spectral clustering, unlike Tikhonov where the bound degrades as $1/\min(\lambda_i + \gamma)$ for clustered eigenvalues.

11. Conclusion

We have presented Differential Spectral Damping (DSD), a regularization formula for ill-conditioned kernel matrix inversion that adapts its penalty to local eigengap structure. Derived from Davis-Kahan perturbation theory, DSD preserves reliable eigenvectors (large gaps) while suppressing corrupted ones (small gaps) through a smooth exponential transition governed by two interpretable parameters.

The primary contribution is classification accuracy on high-dimensional

LSTSVM problems: +4.8pp on GINA (d=970, Cohen's $d = 4.49$), +10.4pp at d=200, and +2.6pp on Madelon (d=500)—all over optimally-tuned Tikhonov with $p < 0.0001$, achieved using only principled initialization without any optimization or cross-validation. The advantage arises from a structural prior: the gap-adaptive initialization encodes spectral reliability information that scalar regularization cannot represent and that gradient optimization on proxy losses actively destroys.

DSD does not improve pre-image reconstruction over Tikhonov (they are equivalent), and provides no benefit below $d \approx 100$ where spectral clustering is insufficient. It is a specialized tool for the high-dimensional, ill-conditioned regime—precisely where kernel methods are most needed and where standard regularization is most inadequate.

The method is computationally free ($O(m)$ overhead over the required eigendecomposition), requires no cross-validation, introduces only two interpretable parameters with principled initialization, and applies to any kernel method whose system matrices exhibit spectral tail-clustering.

Reproducibility

Complete source code (Python/PyTorch), all experiment scripts, and raw results will be made publicly available upon acceptance.

References

[1] C. Cortes, V. Vapnik, Support-vector networks, Machine Learning 20 (3) (1995) 273–297.

[2] T. Hofmann, B. Schölkopf, A. J. Smola, Kernel methods in machine learning, The Annals of Statistics 36 (3) (2008) 1171–1220.

[3] M. A. Kumar, M. Gopal, Least squares twin support vector machines for pattern classification, Expert Systems with Applications 36 (4) (2009) 7535–7543.

[4] M. Tanveer, T. Rajani, R. Rastogi, Y. H. Shao, M. A. Ganaie, Comprehensive review on twin support vector machines, Annals of Operations Research 318 (2022) 1223–1268.

[5] C. Davis, W. M. Kahan, The rotation of eigenvectors by a perturbation. III, SIAM Journal on Numerical Analysis 7 (1) (1970) 1–46.

[6] C. K. I. Williams, M. Seeger, Using the nyström method to speed up kernel machines, in: Advances in Neural Information Processing Systems (NeurIPS), 2001, pp. 682–688.

[7] J. T.-Y. Kwok, I. W.-H. Tsang, The pre-image problem in kernel methods, IEEE Transactions on Neural Networks 15 (6) (2004) 1517–1525.

[8] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: International Conference on Learning Representations (ICLR), 2015.

[9] A. N. Tikhonov, On the solution of ill-posed problems and the method of regularization, Doklady Akademii Nauk SSSR 151 (3) (1963) 501–504.

[10] G. Wahba, Practical approximate solutions to linear operator equations when the data are noisy, SIAM Journal on Numerical Analysis 14 (4) (1977) 651–667.

[11] G. H. Golub, C. F. Van Loan, Matrix Computations, 4th Edition, Johns Hopkins University Press, 2013.

[12] H. W. Engl, M. Hanke, A. Neubauer, Regularization of Inverse Problems, Springer, 1996.

[13] Y.-H. Shao, C.-H. Zhang, X.-B. Wang, N.-Y. Deng, Improvements on twin support vector machines, IEEE Transactions on Neural Networks 22 (6) (2011) 962–968.

[14] S. Mika, B. Schölkopf, A. Smola, K.-R. Müller, M. Scholz, G. Rätsch, Kernel PCA and de-noising in feature spaces, in: Advances in Neural Information Processing Systems (NeurIPS), 1999, pp. 536–542.

[15] B. Schölkopf, S. Mika, C. J. C. Burges, P. Knirsch, K.-R. Müller, G. Rätsch, A. J. Smola, Input space versus feature space in kernel-based methods, IEEE Transactions on Neural Networks 10 (5) (1999) 1000–1017.

[16] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems (NeurIPS), 2017, pp. 4765–4774.

[17] M. T. Ribeiro, S. Singh, C. Guestrin, "Why Should I Trust You?": Explaining the predictions of any classifier, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM, 2016, pp. 1135–1144.

[18] J. Cohen, Statistical Power Analysis for the Behavioral Sciences, 2nd Edition, Lawrence Erlbaum Associates, 1988.

[19] O. Bousquet, A. Elisseeff, Stability and generalization, Journal of Machine Learning Research 2 (2002) 499–526.

Table 1: Symbol Definitions

Symbol	Definition
Data and Dimensions	
n	Number of training samples
d	Input space dimensionality
m	Number of Nyström landmarks or matrix dimension
$\boldsymbol{x}_i \in \mathbb{R}^d$	The i-th training sample
$X \in \mathbb{R}^{n \times d}$	Training data matrix
$y_i \in \{+1, -1\}$	Class label of sample i
Kernel and Feature Spaces	
$\mathcal{H}$	Reproducing Kernel Hilbert Space (RKHS)
$\phi : \mathbb{R}^d \to \mathcal{H}$	Feature map into RKHS
$\kappa(\boldsymbol{x}, \boldsymbol{x}')$	Kernel function: $\kappa(\boldsymbol{x}, \boldsymbol{x}') = \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \rangle_{\mathcal{H}}$
γ_k	RBF kernel bandwidth: $\kappa(\boldsymbol{x}, \boldsymbol{x}') = \exp(-\gamma_k \|\boldsymbol{x} - \boldsymbol{x}'\|^2)$
$\boldsymbol{K} \in \mathbb{R}^{n \times n}$	Full kernel (Gram) matrix: $K_{ij} = \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j)$
Nyström Approximation	
$L = \{\boldsymbol{l}_1, \ldots, \boldsymbol{l}_m\}$	Nyström landmark points ($L \subset X$)
$\boldsymbol{W} \in \mathbb{R}^{m \times m}$	Kernel submatrix: $W_{ij} = \kappa(\boldsymbol{l}_i, \boldsymbol{l}_j)$

Table 2: Regularized Inverse Formulas in the Eigenvalue Domain

Method	$\tilde{\lambda}_i^{-1}$	**Adapts to**
Naive inverse	$1/\lambda_i$	Nothing
Tikhonov	$1/(\lambda_i + \gamma)$	Magnitude only
Truncated SVD	$1/\lambda_i$ if $i > k$, else 0	Rank (hard)
Spectral filter	$g(\lambda_i)/\lambda_i$	Magnitude (soft)
DSD (ours)	$\lambda_i/(\lambda_i^2 + \alpha e^{-\beta\delta_i})$	**Eigengap (reliability)**

Table 3: LSTSVM Classification Accuracy: DSD (auto-init, no tuning) vs. Tikhonov (15-point grid search) vs. Truncated SVD (rank grid search). All differences with $\Delta > 0$ are significant at $p < 0.0001$.

Dataset	**DSD**	**Tik**	**TSVD**	$\Delta_{\textbf{Tik}}$	**Wins**	**C.**d
GINA (d=970)	**85.9**	81.1	83.4	+4.8	30/30	4.49
Madelon (d=500)	**57.7**	55.0	—	+2.6	27/30	1.76
Synth. (d=200)	**71.4**	61.1	—	+10.4	44/50	1.57
Synth. (d=100)	**79.1**	75.8	—	+3.3	35/50	0.39
Synth. (d=50)	90.1	90.1	—	0	25/50	0.0

C.d = Cohen's d. Δ_{Tik} = DSD advantage over Tikhonov (pp).

GINA/Madelon: 30 seeds; synthetic: 50 seeds. Accuracy in %.

TSVD on GINA: 83.4% (DSD wins 30/30, $p < 0.0001$).

Table 4: Kernel-LSTSVM: DSD-optimized vs. Tikhonov-optimized (50 seeds)

Dataset	**DSD (%)**	**Tik (%)**	Δ	p**-value**
Two Moons (γ_k=2.0)	**92.7**	92.1	+0.6pp	0.028
Genomics (d=100)	87.7	87.7	0	0.77

Table 5: Pre-Image Mean Reconstruction Error (Swiss Roll, 50 seeds). DSD and Tikhonov both receive Adam optimization on reconstruction loss.

Noise σ	**DSD-opt**	**Tik-opt**	**Naive**	**DSD vs Tik**
5×10^{-3}	0.069	0.069	4.56	Tie (p=0.99)
10^{-3}	0.051	0.048	0.84	Tik wins (p=0.02)
10^{-4}	0.042	0.035	0.09	Tik wins (p<0.001)

Table 6: Below the Operating Boundary ($d \leq 64$): DSD Not Expected to Win

Dataset	d	**DSD (%)**	**Tik (%)**	Δ	**Result**
Digits	64	88.3	89.1	−0.9pp	Tik wins
Ionosphere	34	73.3	80.3	−7.0pp	Tik wins

Table 7: Spectral Tail-Clustering Severity in LSTSVM Product Matrices

d	**Condition No.**	**Tail Span***	**DSD Δ**
200	8.5×10^3	7.1%	+10.4pp
100	3.5×10^3	11.1%	+3.3pp
50	1.1×10^3	15.1%	0pp

*Spectral range fraction of bottom 50% eigenvalues.

Table 8: Deployment Guidance: When to Use DSD

Condition	**DSD Δ**	**Recommendation**
LSTSVM, $d \geq 100$	+3–10pp	**Use DSD**
Kernel-LSTSVM, sharp RBF	+0.6pp	**Use DSD**
Pre-image (any noise)	≈ Tik	Either method
LSTSVM, $d \leq 50$	0	Simplest method
Low-d real-world	Mixed	Benchmark both